

```

1 from machine import Pin, I2C
2 import time
3
4 i2c = I2C(0, scl=Pin(22), sda=Pin(21))
5 addr = 0x48
6
7 # kalibrace
8 gain = 1.00
9 offset = 0.08
10
11 # korekce nad 6V
12 korekce = 0.98
13
14
15 def read_adc_A0():
16     config = bytearray(2)
17     config[0] = 0xC2
18     config[1] = 0x83
19
20     i2c.writeto_mem(addr, 0x01, config)
21     time.sleep(0.1)
22
23     data = i2c.readfrom_mem(addr, 0x00, 2)
24
25     raw = (data[0] << 8) | data[1]
26
27     if raw > 0x7FFF:
28         raw -= 0x10000
29
30     return raw
31
32
33 while True:
34     raw = read_adc_A0()
35
36     # napětí na vstupu ADS1115
37     v_adc = raw * 0.000125
38
39     # základní kalibrace
40     v_in = (v_adc * 4) * gain - offset
41
42     # rozdělení rozsahu
43     if v_in > 6:
44         v_in = v_in * korekce
45
46     # ochrana
47     if v_in < 0:
48         v_in = 0
49
50     print(round(v_in, 2))
51
52     time.sleep(0.5)
53

```